

Question	Answer	Marks
5(a)(i)	time for one oscillation / vibration / cycle (of a particle / point on wave) or time between <u>adjacent</u> wavefronts (passing the same point) or <u>shortest</u> time between two wavefronts (passing the same point)	B1
5(a)(ii)	(The magnitude of the) maximum displacement (of a particle / point on the wave)	B1
5(b)(i)	a cross vertically in line with a peak or trough on the string	B1
5(b)(ii)	$(\lambda =) 0.96 \times 2 / 3$ $= 0.64$	M1
	$v = f\lambda$	C1
	$= 25 \times 0.96 \times 2 / 3 = 16 \text{ (m s}^{-1}\text{)}$	A1
5(b)(iii)	$\lambda = 16 / 30$ or $\lambda = 0.53$ or $\lambda = L / 1.8$ or $\lambda = 0.55L$	B1
	$\lambda \neq 2L / n$ or $\lambda \neq 1.92 / n$ <u>and</u> (so) stationary wave not formed	B1
	OR $f \neq vn / 2L$ or $f \neq 16n / 2L$ or $f \neq 8.3n$ <u>and</u> (so) stationary wave not formed	B2
5(c)(i)	$n = d \sin \theta / \lambda$	C1
	$= ((1 / 6.7 \times 10^5) \times \sin(37)) / 4.5 \times 10^{-7}$	C1
	$= 2$	A1
5(c)(ii)	decreases	B1